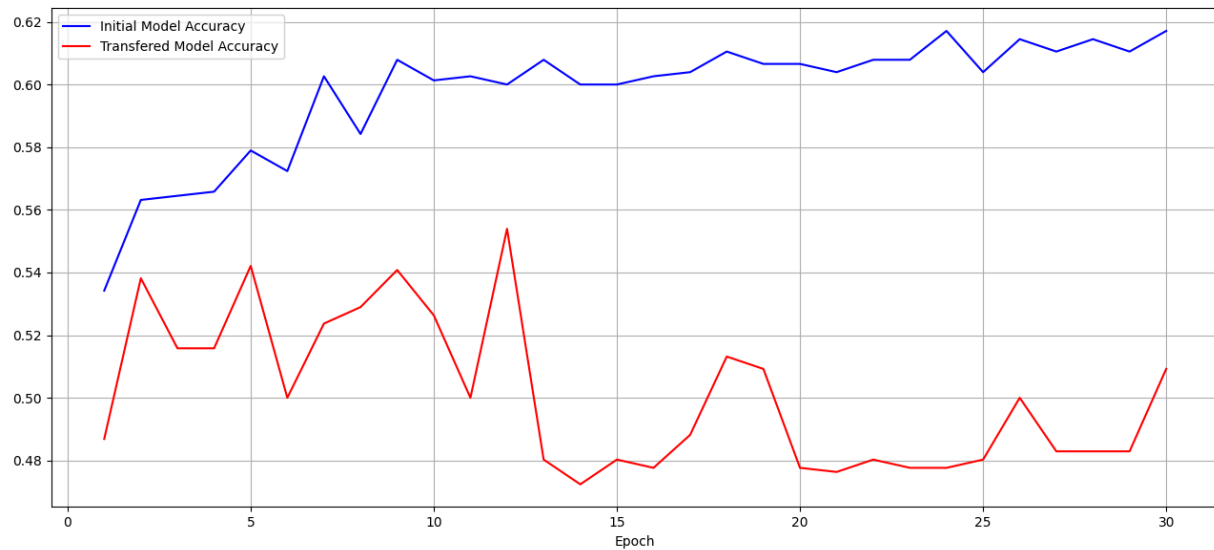


Transfer Learning Report

Far Transfer from Facial Recognition to Dog vs Cat recognition



The blue curve represents accuracy of with transfer and the red curve represents accuracy of without transfer (We think the naming on the graph legend was incorrect in the transfer_data.py file)

Final Classification Accuracy on Test Set:

With transfer

* Evaluating transfered_model

4/4 [=====] - 0s 57ms/step - loss: 0.6318 - accuracy: 0.6343

Without transfer

* Evaluating random_model

4/4 [=====] - 0s 58ms/step - loss: 0.6869 - accuracy: 0.5174

Facial Recognition Classes images

Happy: 

Neutral: 

Surprised: 

Dog and Cat Classes images



Model Summaries

With Transfer:

* Training transfered_model for 30 epochs

Model: "model"

Layer (type)	Output Shape	Param #
rescaling_input (InputLayer)	[(None, 150, 150, 3)]	0
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 147, 147, 16)	784
max_pooling2d (MaxPooling2D)	(None, 36, 36, 16)	0
dropout (Dropout)	(None, 36, 36, 16)	0
conv2d_1 (Conv2D)	(None, 33, 33, 32)	8224
max_pooling2d_1 (MaxPooling2D)	(None, 11, 11, 32)	0
dropout_1 (Dropout)	(None, 11, 11, 32)	0
conv2d_2 (Conv2D)	(None, 8, 8, 48)	24624

max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 48)	0
dropout_2 (Dropout)	(None, 4, 4, 48)	0
flatten (Flatten)	(None, 768)	0
dense (Dense)	(None, 64)	49216
dropout_3 (Dropout)	(None, 64)	0
transfer_dense (Dense)	(None, 32)	2080
transfer_output (Dense)	(None, 2)	66

Total params: 84,994

Trainable params: 2,146

Non-trainable params: 82,848

Without transfer:

* Training random_model for 30 epochs

Model: "model_1"

Layer (type)	Output Shape	Param #
rescaling_input (InputLayer)	[(None, 150, 150, 3)]	0
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 147, 147, 16)	784
max_pooling2d (MaxPooling2D)	(None, 36, 36, 16)	0
dropout (Dropout)	(None, 36, 36, 16)	0
conv2d_1 (Conv2D)	(None, 33, 33, 32)	8224
max_pooling2d_1 (MaxPooling2D)	(None, 11, 11, 32)	0
dropout_1 (Dropout)	(None, 11, 11, 32)	0

conv2d_2 (Conv2D)	(None, 8, 8, 48)	24624
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 48)	0
dropout_2 (Dropout)	(None, 4, 4, 48)	0
flatten (Flatten)	(None, 768)	0
dense (Dense)	(None, 64)	49216
dropout_3 (Dropout)	(None, 64)	0
transfer_dense (Dense)	(None, 32)	2080
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Total params: 84,994

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